

PRAKTIKUM BASIS DATA

MODUL 11

NoSQL – Key Value Database



Oleh:

Tony Hendrawan 103122400021

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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Jurnal

1. Buat Keyspace

Output:

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing. Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifine_0801
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE spotifine_0801;
cqlsh:spotifine_0801> |
```

2. Buat Table

Output:

```
cqlsh:spotifine_0801> CREATE TABLE musician_data (
...     artist_id uuid PRIMARY KEY,
...     artist_name text,
...     followers int,
...     origin text,
...     genres set<text>
... );
cqlsh:spotifine_0801> INSERT INTO musician_data (artist_id, artist_name, followers, origin, genres)
... VALUES (uuid(), 'Tulus', 5000000, 'Indonesia', {'Pop', 'Jazz'});
cqlsh:spotifine_0801>
cqlsh:spotifine_0801> INSERT INTO musician_data (artist_id, artist_name, followers, origin, genres)
... VALUES (uuid(), 'The Weeknd', 85000000, 'Canada', {'R&B', 'Pop', 'Synth-pop'});
cqlsh:spotifine_0801>
cqlsh:spotifine_0801> INSERT INTO musician_data (artist_id, artist_name, followers, origin, genres)
... VALUES (uuid(), 'Tompi', 7600000, 'Canada', {'Pop', 'Synth-pop'});
cqlsh:spotifine_0801> SELECT * FROM musician_data;

artist_id | artist_name | followers | genres | origin
-----|-----|-----|-----|-----
0ae19d22-5b52-4961-86bf-e10aefb01497 | Tulus | 5000000 | {'Jazz', 'Pop'} | Indonesia
6dade861-328d-446e-960f-8dddc1d97dc2 | Tompi | 7600000 | {'Pop', 'Synth-pop'} | Canada
a6f3daaa-b58b-4e53-869f-3e87379f785c | The Weeknd | 85000000 | {'Pop', 'R&B', 'Synth-pop'} | Canada

(3 rows)
cqlsh:spotifine_0801> |
```

3. Apa keuntungan utama menggunakan tipe data set pada kolom genres di aplikasi Spotifine dibandingkan jika ia menggunakan tabel relasional (RDBMS) biasa?

Jawaban:

performa tinggi melalui denormalisasi, di mana Cassandra menghilangkan kebutuhan operasi JOIN yang berat pada RDBMS dengan menyimpan semua genre langsung di dalam satu baris data musisi.